CYBER SECURITY 1 Cyber Security Internship – Task 1 Vulnerability Assessment Report

Name: Tirumani Divya Sri
Role: Cyber Security Intern
Organization: Future Interns

Introduction

As part of my Cyber Security Internship, I performed a read-only security assessment of a public website to identify common security weaknesses and present risks in a professional format.

Target Website: http://testphp.vulnweb.com

Scope:

•Public pages only
•Passive analysis only
•No exploitation performed

Tools Used :

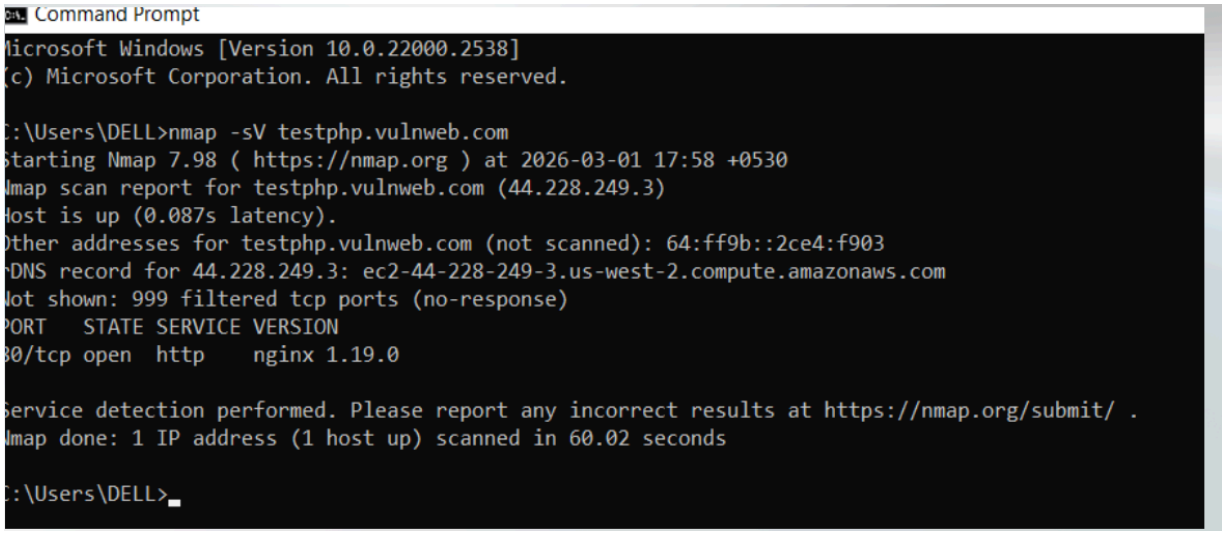
Nmap
Security Header Scanner
Browser Developer

Tools & Evidence

Figure 1: Nmap scan showing open port 80 and nginx version
The Nmap scan shows port 80 open and nginx version visible.

Exposing server details can help attackers identify known vulnerabilities.

Risk Level: Medium



```
Command Prompt
Microsoft Windows [Version 10.0.22000.2538]
(c) Microsoft Corporation. All rights reserved.

C:\Users\DELL>nmap -sV testphp.vulnweb.com
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-01 17:58 +0530
Nmap scan report for testphp.vulnweb.com (44.228.249.3)
Host is up (0.087s latency).
Other addresses for testphp.vulnweb.com (not scanned): 64:ff9b::2ce4:f903
rDNS record for 44.228.249.3: ec2-44-228-249-3.us-west-2.compute.amazonaws.com
Not shown: 999 filtered tcp ports (no-response)
PORT   STATE SERVICE VERSION
80/tcp open  http    nginx 1.19.0

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 60.02 seconds

C:\Users\DELL>
```

Figure 2: Missing security headers detected
Important security headers such as Content-Security-Policy and X Frame-Options are missing. This reduces protection against browser-based attacks**.**

Risk Level: Medium

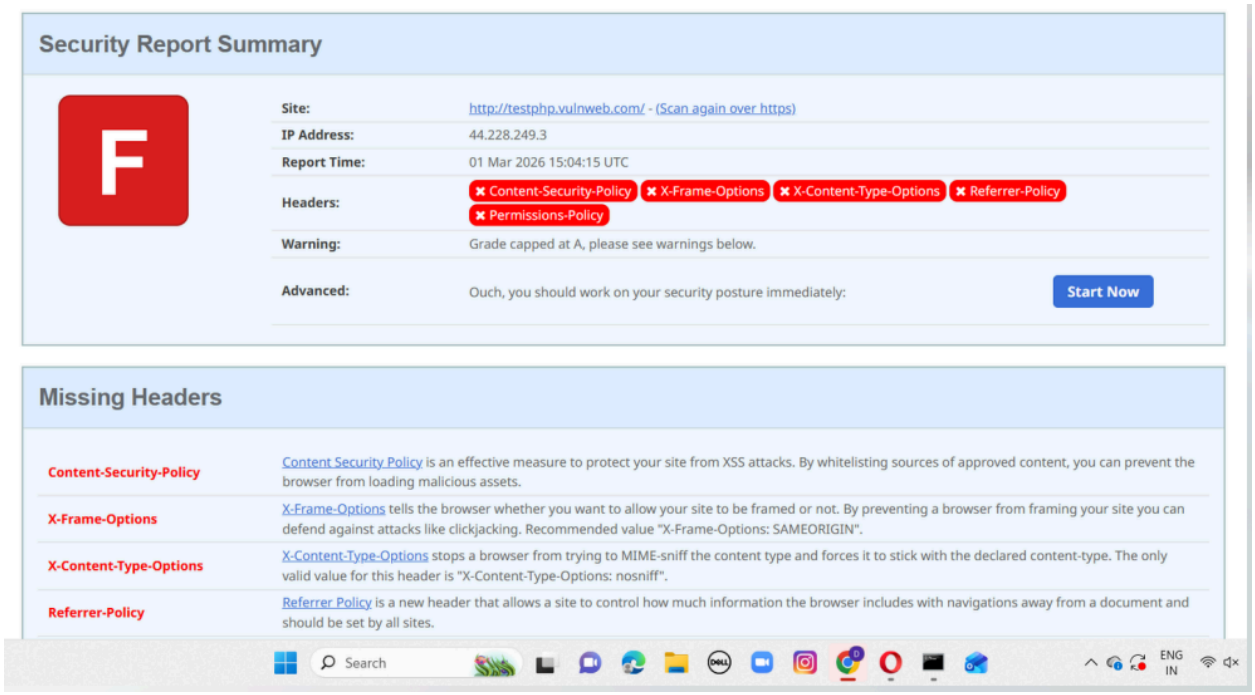


Figure 3: Website accessible over HTTP
The website is accessible over HTTP and does not redirect to HTTPS. This means communication is not encrypted and could be intercepted.

Risk Level: High

Risk Assessment

The absence of HTTPS was classified as High risk. Missing headers and exposed server information were classified as Medium risk

Conclusion:

This task provided hands-on experience in ethical vulnerability assessment, risk classification, and professional security reporting.

Thank You Cyber Security Internship – 2026